

The central concept to understand is the Amazon Machine Image (AMI).
AMI = master blueprint/copy of a server's root drive.
It's a template that contains:

- The Operating System (e.g., Linux, Windows Server).

- Any pre-installed software and applications (e.g., a web server, a database, your custom code).

- The specific configurations and permissions for the instance.

What happens when we "launch instance" aka VM Provisioning in AWS

Step 1: The Launch Request (API Call)

An API call is made to the EC2 Control Plane

Consider the EC2 Control Plane as a massive, distributed "brain" of the EC2 (Elastic Compute Cloud) service.

Our request contains the specifics/requirements:

- AMI ID,
- Instance type (CPU and RAM),
- Boot disk/hard drive type - say SSD and Boot disk size - say 100 GB
- Additional volumes if needed
- Networking settings (VPCs, Subnets) and
- security groups (inbound/outbound)

Step 2: Finding a Home (Where should the VM be created or placed)

The EC2 Control Plane finds a suitable physical server (a host machine) in the selected Availability Zone that has enough available CPU and RAM (satisfying the requirements we specified)

Step 3: Resource Allocation on the Host

Once a host is selected, the AWS Nitro Hypervisor (or any other hypervisor) on that physical machine gets the instruction to provision the VM - carving out/allocating the specification (number of vCPUs & required RAM from that single physical server).

Step 4: Creating the "Hard Drive" (The Critical Step)

The AMI we selected contains a reference to a snapshot of a root volume, which is stored in Amazon S3 (we will cover this service today during the contact session). Essentially we can regard this as an upload from the storage!!!!

EC2 Control Plane makes an internal call to EBS Control Plane to do the needed task

A brand new Elastic Block Store (EBS) volume is created from that snapshot.

This new EBS volume is an exact, block-for-block clone of the master template.

This new EBS volume is the "boot disk" or "hard drive" for the new virtual machine.

Point to ponder:

What is this EBS service?

It is a massive distributed storage system. The data for the single EBS volume is not stored on a single 100 GB physical SSD. Instead, it is:

Chunked: Broken down into smaller pieces (chunks or blocks).

Replicated: Each chunk is copied multiple times for redundancy.

Distributed: These replicated chunks are spread across many different physical hard drives in many different server racks, all within the same Availability Zone.

Step 5: Network Attachment and Boot-Up

The newly created EBS volume is network-attached to the newly provisioned VM on the host server.

Is it physically inside the box - No! It is connected over AWS's high-speed internal network.

An Elastic Network Interface (ENI) is created and attached to the VM, giving it a private IP address and connecting it to the chosen Virtual Private Cloud (VPC).

The hypervisor then instructs the VM to boot from its attached EBS volume. The Operating System starts up, just like a physical computer booting from its hard drive.

Step 6: Health Checks and Availability

The EC2 service runs health checks on the new instance. Once it passes these checks and the OS is fully booted, the instance is marked as "running" in the AWS Web console.

We can connect to it via SSH/RDP/AWS Instance Connect.

The entire process, from the API call to a running state, often takes less than a minute.

Analogy: Ordering a Custom-Built PC online

Launch Configuration (AMI, instance type): This is the order form or spec sheet ("We state that we want this motherboard, this OS, and this pre-installed software").

The EC2 Control Plane: This is the factory's automated management system that receives the order and assigns it to an assembly line.

The Physical Host Server: This is the workbench and the technician who will build the PC.

The EBS Snapshot: This is the factory's "master hard drive image" containing a perfect installation of the OS we needed.

The EBS Volume: The technician takes a blank hard drive and clones the master image

onto it. This new drive is the personal hard drive.

The Final Product (EC2 Instance): The technician assembles the PC at the workbench, plugs in the newly cloned hard drive, and powers it on.

This model is what makes the cloud so powerful: we can create thousands of identical, perfectly configured servers from a single master template (the AMI) in minutes.